Fiscal Year:	FY 2026	**Task Last Updated:**	FY 03/05/2026
PI Name:	Chiow, Sheng-Wey Ph.D.		
Project Title:	Direct Detection of Dark Energy on Einstein Elevator (D3E3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91109	**Congressional District:**	27
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	02/25/2021	**End Date:**	09/30/2028
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2028 per D. Griffin/NASA-HQ. The original period of performance was 02/25/21 - 09/30/2026 (Ed., 1/21/25).		
Key Personnel Changes/Previous PI:	Per the Principal Investigator, Nan Yu is a Co-Investigator on the project. Dr. Yu is with the NASA Jet Propulsion Laboratory (Ed., 5/2/25).		
COI Name (Institution):	Yu, Nan (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15884		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Technical Background
Dark energy is the greatest mystery in fundamental physics to date. 70% of the Universe energy content is in the form of dark energy, which is responsible for the observed accelerated expansion. While little is known about the nature of dark energy, it is conjectured to be a new scalar field that weakly interacts with normal matter. To be consistent with the current solar system observations, however, such interactions should be environment dependent and screened locally. Possible scalar fields with screening mechanisms are described in the forms such as chameleon, symmetron, and galileon models. Recently, cold atom experiments using atom interferometers in a ground laboratory have contributed significantly to the constraints of chameleon and symmetron models, thanks to the “thin-shell” effect of these two models.

Chameleon and symmetron models emerged from the cosmology community, where the above-mentioned screening mechanism is realized, in the framework of the scalar field theory. By incorporating these models in the master equations of atomic sensors, which utilize quantum mechanical properties of individual atoms for force measurements, theory predicts that there must be extra forces merely due to the presence of dark energy fields, and that these forces are orders of magnitude weaker than gravity. We propose to operate atomic sensors near a structured mass as the source of dark energy force in microgravity. Microgravity allows extended interrogation time and thus enhanced sensitivity in a small and well-characterized package, and the structured source mass is designed to suppress the gravity effect while maintaining the dark energy signal. These innovative approaches enable sensitivity beyond those achieved in laboratory experiments.

Programmatic Background

As one of our key international partners, collaborative research with the German Aerospace Center (DLR) into direct detection of dark energy in a DLR facility is an excellent programmatic rationale to further build our relationship with DLR, exchange technical knowledge with DLR, and develop and demonstrate critical atom interferometry technology, in the pursuit of this proposed directed research activity.

As a result of the 2011 NRA (NASA Research Announcement), NASA selected Holger Muller (Berkeley) and Nan Yu to participate in the European Space Agency (ESA) Quantum Weak Equivalence Principle (QWEP) study from 2012 to 2017. As part of that study, they realized they could use the QWEP atom interferometer approach to search for the chameleon dark energy candidate.

Muller and his students designed an apparatus and performed a dark energy experiment. In addition to the NASA support, this activity was also supported by National Science Foundation (NSF) and Defense Advanced Research Projects Agency (DARPA). They published their results in Science in 2015.

In a 2018 Phys Rev D paper extending the previous work, Sheng-wey Chiow and Nan Yu noted that, “For pointlike particles such as atoms, the depth of screening is larger than the size of the particle, such that the screening mechanism is ineffective and the chameleon force is fully expressed on the atomic test particles. Extra force measurements using atom interferometry are thus much more sensitive than bulk mass based measurements, and indeed have placed the most stringent constraints on the parameters characterizing chameleon field.” They further wrote that, “In this paper, we present a measurement concept of direct detection of chameleon forces using atom interferometers in microgravity, with a sensitivity sufficient to detect any predicted chameleon force or rule out the theory completely.”

While the only precision measurements with atom interferometers to date have only been demonstrated in ground laboratories, there have been significant efforts recently to bring the technology to a microgravity environment and space, including drop towers, sounding rocket experiments, and the Cold Atom Laboratory. However, none of these experiments will demonstrate precision measurements beyond the state of the art on the ground. Even Bose-Einstein Condensate and Cold Atom Laboratory (BECCAL) will not have true precision measurement capabilities per its baseline plan approved at the Preliminary Design Review. On the other hand, the University of Hannover Einstein Elevator (EE) microgravity facility, with high dropping repetition rate of 300 per day, offers the possibility for accumulating enough data through averaging for precision measurements. With the commissioning of EE and the availability of the German DLR MAIUS-I (Matter-Wave Interferometry in Microgravity) BEC (Bose-Einstein condensate) machine, an opportunity arises for conducting the first precision measurements in a microgravity environment. Given the JPL Planning, Programming, Budgeting, and Execution (PPBE) line to develop atom interferometry (AI) for the Einstein Elevator, NASA can fund developing and verifying a microgravity instrument that could either detect chameleon dark energy particles or rule such a theory out completely. Given the status of modern physics, this is a highly significant effort.

Implementation

We propose to use this opportunity to collaborate with DLR for direct detection of dark energy measurements. The atomic sensor will be implemented in an ultra-high vacuum chamber capable of generating ultra-cold atoms. We plan on leveraging the existing hardware of MAIUS-I, a compact cold atom instrument that was used in a DLR sounding rocket experiment. Professor Ernst Rasel and his team at Hannover University will repurpose MAIUS-I and modify the vacuum chamber for dark energy measurements. We will work with DLR to conduct the experiment in the Einstein Elevator (EE) at Hannover, where 4-s of microgravity time will be available every 5 min. The free-fall time and the repetition rate will provide sufficient sensitivity for pushing the boundaries of dark energy models.

Summary

The proposed D3E3 will improve the state-of-the-art constraints on scalar field theories by several orders of magnitude by performing cold atom experiments in the microgravity environment available in the Einstein Elevator. The improved sensitivity will close the gap in the parameter space of chameleon for dark energy density at currently observed level, decisively testing the validity of the chameleon model.

References:

Hamilton P, Jaffe M, Haslinger P, Simmons Q, Müller H, Khoury J. ASTROPHYSICS. Atom-interferometry constraints on dark energy. Science. 2015 Aug 21;349(6250):849-51. https://

Chiow S, Yu N. "Multiloop atom interferometer measurements of chameleon dark energy in microgravity." Physical Review D. 2018 Feb;97(4):044043. https://

Rationale for HRP Directed Research:	Highly Constrained Research The goal of this research is to verify whether or not Atom Interferometry (AI) can detect or rule out the chameleon theory of dark energy It is constrained by the following: Based on the previous work the Jet Propulsion Laboratory (JPL) has already initiated a collaboration with University of Hannover scientists, Prof. Ernst Rasel and Prof. Wolfgang Ertmer, to perform this research in the Einstein Elevator. They will be able to re-purpose MAIUS (Matter-Wave Interferometry in Microgravity) rocket hardware for use in the elevator, as long as approval is given to move forward on this in the short term. They have submitted a proposal to refurbish the MAIUS laser system for this collaboration to the German Aerospace Center (DLR), and are anticipating a positive answer. If we are unable to take advantage of the availability of the MAIUS hardware for this purpose now, we may lose the opportunity for this research altogether. It is expected that to perform a similar experiment on the International Space Station (ISS) would cost at least as much as the Cold Atom Laboratory (CAL) development, depending on the ultimate requirements chosen for the flight project. Insufficient Time for Solicitation JPL does not have the funding or the opportunity to solicit research from the community in this area. The next NASA Research Announcement (NRA) is years away, and it is specifically targeted for performing research using the as-yet-built DLR funded Bose-Einstein Condensate and Cold Atom Laboratory (BECCAL) follow-on ISS experiment to CAL. The opportunity from the availabilities of Einstein Elevator (EE) and MAIUS-I is opportunistic and unanticipated. The Einstein Elevator is currently in the commissioning phase, and is expected to be fully operational in two years. At the same time, the repurpose of MAIUS-I for EE is already planned. It is anticipated that atom interferometer experiments will have priority for the first few years.
Research Impact/Earth Benefits:	
Task Progress:	The Hannover team conducted four flight campaigns in the Einstein Elevator facility in June, August, November, and December of 2025. The experimental apparatus was launched in the Einstein Elevator for 4-seconds of microgravity time in each experimental run. Issues were identified and mitigated on all subsystems: the integrity of ultra-high vacuum, the stability of the laser optical system, power and electronics, and the water cooling system. The Hannover team repeatably generated a three-dimensional magneto-optic trap (3D-MOT) loaded from a 2D-MOT of rubidium-87 during the flights. This is a crucial step toward generating Bose-Einstein Condensate (BEC) of Rb-87 for atom interferometry demonstration and dark energy model testing. The NASA Jet Propulsion Laboratory (JPL) team has been supporting the Hannover team on hardware development, troubleshooting, and technology advancement. For instance, a scheme of stabilizing relative phase of laser beams of atom interferometry was proposed. To conduct atom interferometry, counter-propagating laser pulses of a few 100-microseconds long from separate optical fibers are to be fired simultaneously at the ultracold cloud of Rb-87 atoms. Due to phase disturbances induced by the optical fibers and other optics, mostly from the shock of launch in the Einstein Elevator, the phase between two laser pulses is randomized, which results in randomized atom interferometer output. While the D3E3 experiment is designed with simultaneous dual atom interferometers to suppress the impact of laser phase noise, it is advantageous to be able to stabilize and control the laser phase, e.g., to be able to controllably scan the interferometer output, and to “park” the interferometer output at the most sensitive phase setting. The short duration of pulses poses a challenge to actively stabilize the laser phase. A scheme with a weak, off-resonant, leading light to facilitate phase locking was discussed. The joint team agreed the feasibility of implementation and is moving forward with a lab demo. Moreover, detailed detection plans were developed. After the atom interferometry, atoms are in the same (internal) state. Time-of-flight is required to allow different output ports of an interferometer to spatially separate and to be imaged on a camera. Tight constraints on available time in microgravity, detection beam size, camera size, separation velocity, etc., manifested as a multi-dimensional optimization problem. By carefully evaluating operation scenarios and hardware limitations, the team concluded that the detection capability will not impede the data collection. A summary paper of the joint project was developed by the team members. It was peer-reviewed and published in European Physics Journal, Special Issue on Quantum Technology. [Ed. Note: See Bibliography.] In addition, science implication investigations were focused on refining scalar field constraint expectations based on the source mass geometry and anticipated experimental parameters. In light of the challenge on the standard model of cosmology (ΛCDM) by DESI Data Release 2 (DR2), where observations on millions of galaxies do not agree with the ΛCDM prediction, the team identified that chameleon-like dark energy model fits one of the mitigation proposals. This finding strengthens the scientific value of D3E3. A detailed analysis on the impact of finite laser-pulse duration on atom interferometry was conducted. Typically, researchers treat the laser pulses in an atom interferometer as infinitely short. The correction to this treatment was discussed and formulated in literature. The team took a systematic approach to this problem and considered pulses with non-trivial temporal profiles. Closed-form expressions were obtained for corrections. The result and its applicability were presented in SPIE Quantum West, 2025. In parallel, the need of ultracold high-flex atom source technology for precision quantum sensing was identified. In short, the sensitivity of an ideal atom interferometer is limited by the number of participating atoms, and a long-interrogation time atom interferometer calls for extremely low cloud temperatures to limit the thermal expansion. We surveyed state-of-the-art atom cooling techniques, and have begun lab demonstration efforts for technology advancement. We will continue working with the Hannover/DLR partners on experiment payload setup, microgravity campaigns, troubleshooting, technology maturation, onsite visit and in-depth technical discussions, conference presentations, and journal publications.
Bibliography Type:	Description: (Last Updated: 03/16/2026)

Articles in Peer-reviewed Journals	Garcion C, Gill S, Misslisch M, Heidt A, Papadakis I, Piest B, Schkolnik V, Wendrich T, Prat A, Bleeke K, Müntinga H, Krutzik M, Chiow S, Yu N, Lotz C, Gaaloul N, Rasel EM. "Dark energy search by atom interferometry in the Einstein-elevator." EPJ Quantum Technol. 2025 Dec;12(1):1-21. https://doi.org/10.1140/epjqt/s40507-025-00371-0 , Dec-2025
Conference Materials (Downloadable)	Chiow S, Yu N, Garcion C, Gill S, Misslisch M, Rasel EM, Papadakis I, Krutzik M. "Dark energy detection at the Einstein-Elevator using atom interferometry." SPIE Quantum West 2025 (International Society for Optics and Photonics), San Francisco, CA, January 28-30, 2025. , Jan-2025 SPIEQW_D3E3_2025_v1.pdf